

$$\frac{B}{X} > \frac{C}{W_{ici.}}$$

1. Data parallelism: activations sharded along batch, parameters and optimizer state are replicated on each device. Communication only occurs during the backwards pass.

$$\text{In}[B_X, D] \cdot_D W_{\text{in}}[D, F] \cdot_F W_{\text{out}}[F, D] \rightarrow \text{Out}[B_X, D]$$

2. Fully-sharded data parallelism (FSDP or ZeRO-3): activations sharded along batch (like pure data parallelism), parameters sharded along same mesh axis and AllGathered just-in-time before use in forward pass. Optimizer state also sharded along batch. Reduces duplicated memory.

$$\rightarrow \text{In}[B_X, D] \cdot_D W_{\text{in}}[D_X, F] \cdot_F W_{\text{out}}[F, D_X] \rightarrow \text{Out}[B_X, D]$$

different in ops.

3. Tensor parallelism (also called Megatron sharding or model parallelism): activations sharded along D (d_{model}), parameters sharded along F (d_{ff}). AllGather and ReduceScatter activations before and after each block. Compatible with FSDP.

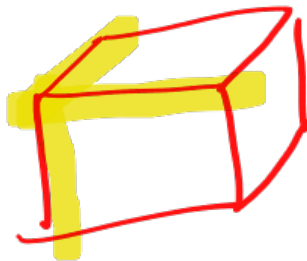
$$\text{In}[B, D_Y] \cdot_D W_{\text{in}}[D, F_Y] \cdot_F W_{\text{out}}[F_Y, D] \rightarrow \text{Out}[B, D_Y]$$

Col-wise parallel → Row sharding

Column + row reduce scatter instead of all-reduce

4. Pipeline parallelism: weights sharded along the layer dimension, activations microbatched and rolled along the layer dimension. Communication between pipeline stages is minimal (just moving activations over a single hop). To abuse notation:

$$\text{In}[L_Z, B, D][i] \cdot_D W_{\text{in}}[L_Z, D, F][i] \cdot_F W_{\text{out}}[L_Z, F, D][i] \rightarrow \text{Out}[L_Z, B, D][i]$$



$$\frac{B}{X} > \frac{C}{W_{ici.}} \times 3.$$

TPU v5p architecture

$$\frac{4.6e^{14}}{1.3e^{11}} \approx 2550.$$

$$\frac{B}{X} > 2550.$$

per device

$$= \boxed{B > 2550}$$

$$\boxed{B > 2550}$$

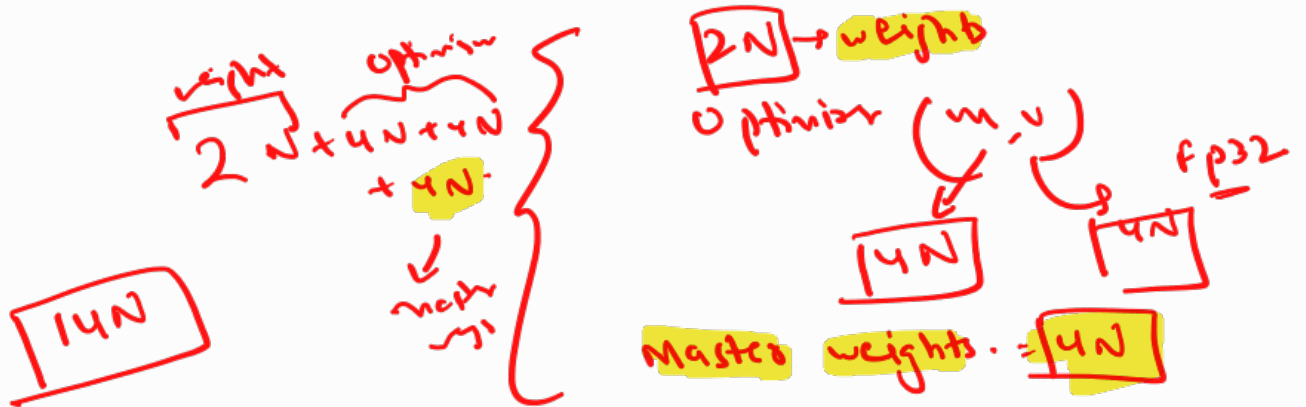
compute bound.

$$\boxed{250 \times 16 \times 20 \times 77}$$

might not be enough

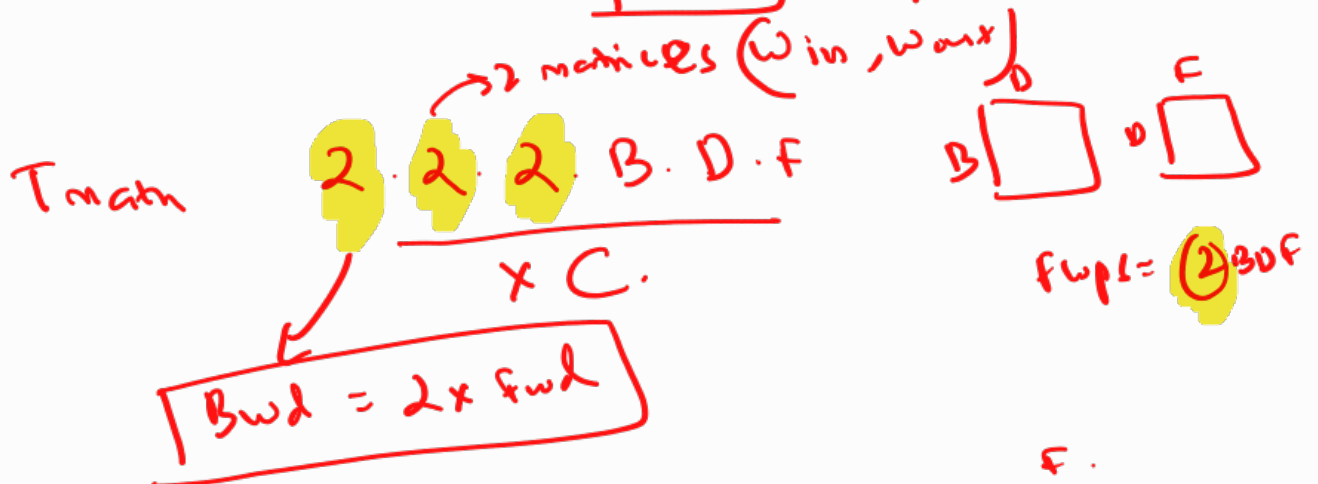
Disadvantage of DP

N parameters. $\rightarrow$ bf16 $\rightarrow$ 2



4 Billion per rank

$14 \times 4 = 56 \text{ GB} \rightarrow$ per rank.



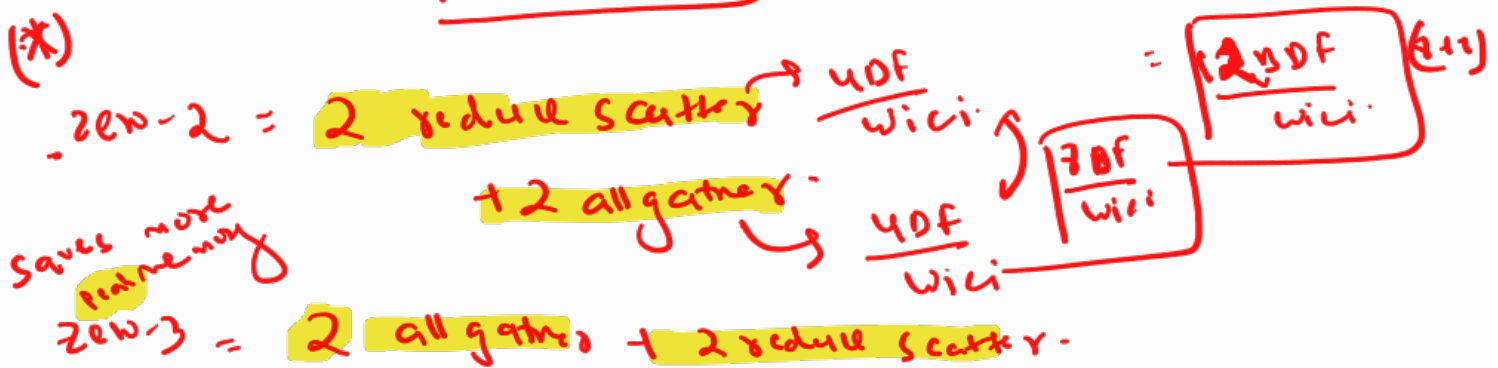
All gather

All reduce = $2 \times$ All gather.

DDP $\rightarrow$ 2 all reduce

zero-1 = 2 all reduce

+ 2 all gather



Saves more
memory

zero-3 = 2 all gather + 2 reduce scatter.

$fwd = 2 \text{ math ops}$
 $\frac{2BDF}{X}$ or $2B \times DF$

Strategy	Compute per layer (flops) (ignoring gating einsum)	Comms per layer (bytes, forward + backward pass)
DP	$4BDF/X + 8BDF/X$ $2 \times fwd$	$0 + 8DF$ $\rightarrow$ all reduce $= 2 \times$ allgather
FSDP	$4BDF/X + 8BDF/X$	$4DF + 8DF$
TP	$4BDF/Y + 8BDF/Y$ $2 \times$ allgather	$4BD + 4BD \rightarrow 2 \times$ reduce scatter
FSDP + TP	$4BDF/(XY) + 8BDF/(XY)$	$(4BD/X + 4DF/Y) + (8BD/X + 8DF/Y)$

$4BDF/X$

fwd
 $4DF + 8DF$
 $2 \times$ allgather
 $2DF$
 bwd
 $2 \times$ reduce scatter
 $2 \times$ allgather

TP

in $[B, D_Y]$, win $[D, f_Y]$, want $[f_Y, D]$
 $\rightarrow$ out $[D, D_Y]$

Communications		Computations	
fwd	bwd	fwd	bwd
allgather $2BD$	$2BD$ $\rightarrow$ allgather	$\frac{2BDF}{Y} + \frac{2BDF}{Y}$	$\frac{4BDF}{Y}$
reduce scatter $2BD$	$2BD$ $2BD$ $2BD$ $(4BD)$	$= \frac{4BDF}{Y}$	$\frac{4BDF}{Y}$

computation bwd

$\frac{BDF}{YC} > \frac{4BD}{wici} = \frac{F}{Y} > \frac{C}{wici}$

Truth

$$\rightarrow \text{DP}_{flop} \quad \frac{B}{X} > \frac{C}{W_{ici}} \quad \frac{B \uparrow}{\text{free lunch.}}$$

$$\rightarrow \text{TP} = \frac{f}{Y} > \frac{C}{W_{ici}} \quad f \uparrow$$



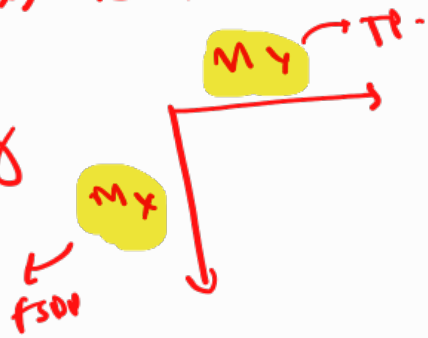
FSOP + TP.

$$\text{In}[B_x, D_y] \cdot \text{Win}[D_x, f_y] \cdot f \cdot \text{Wout}[f_y, D_x] \rightarrow \text{Out}[B_x, D_y]$$

Communication		Computation	
FWD	BWD	fwd	BWD.
$\frac{2BD}{X}$		$\frac{2BDF}{XY}$	
$\frac{2DF}{Y}$		$\frac{2BDF}{XY}$	
$\frac{2DF}{Y}$			
$\frac{2B}{X} \cdot D$			

In $[B_x, D_y]_0$. Win $[D_x, F_y]_f$ won $[F_y, D_x] \rightarrow$ out $[B_x, D_y]$ (forward)

TP $\rightarrow$ activation
FSDP $\rightarrow$ weight & scaling



$$T_{FSDP} = \frac{2 \cdot 2 \cdot D \cdot F}{Y \cdot W_{ici} \cdot M_x}$$

$$T_{TP} = \frac{2 \cdot 2 \cdot B \cdot D}{X \cdot W_{ici} \cdot M_y}$$

max. $\left(\frac{Y \cdot D \cdot F}{Y \cdot W_{ici} \cdot M_x}, \frac{X \cdot B \cdot D}{X \cdot W_{ici} \cdot M_y} \right) \rightarrow T_{mchn} = \frac{2 \cdot 2 \cdot B \cdot D \cdot F}{\frac{N \cdot C}{X \cdot Y}}$

$$\frac{Y \cdot D \cdot F \cdot X}{N \cdot W_{ici} \cdot M_x} \leftarrow \frac{Y \cdot D \cdot F}{Y \cdot W_{ici} \cdot M_x} = \frac{Y \cdot B \cdot D}{X \cdot W_{ici} \cdot M_y} \left\{ \frac{Y \cdot D \cdot F \cdot X}{N \cdot W_{ici} \cdot M_x} = \frac{X \cdot B \cdot D}{X \cdot W_{ici} \cdot M_y} \right.$$

$\hookrightarrow N/X$

$$X^2 = \frac{B \cdot N \cdot M_x}{M_y}$$

$$T_{FSDP} \propto X$$

$$T_{TP} \propto \frac{1}{X}$$

$$X = \sqrt{\frac{B \cdot N \cdot M_x}{F \cdot M_y}}$$

$$X_{opt} = \sqrt{\frac{B \cdot N \cdot M_x}{F \cdot M_y}}$$

$$T_{FSDP} = T_{TP}$$

$\rightarrow T_{mchn}$

$$\frac{B}{N} > \frac{(C/W_{ici})^2}{M_x \cdot M_y \cdot F}$$

output m.

$$\frac{X \cdot B \cdot D \cdot F}{X \cdot Y \cdot C} > \frac{X \cdot D \cdot F}{X \cdot W_{ici} \cdot M_x}$$

$$\frac{B}{X \cdot C} > \frac{1}{W_{ici} \cdot M_x}$$

$$\frac{B}{C/W_{ici}} > \frac{X}{M_x} \Rightarrow \frac{B}{C/W_{ici}} > \sqrt{\frac{B \cdot N \cdot M_x}{F \cdot M_y}}$$

$$\frac{B}{C/W_{ici}} > \sqrt{\frac{B \cdot N}{F \cdot M_x \cdot M_y}}$$

$$= \left(\frac{\sqrt{B F M_x M_y}}{C / \sqrt{N}} \right)^2 (1)^2$$

$\frac{B}{N} > \frac{q^2}{M_x M_y F}$

(*)

Question 1: How many parameters does LLaMA-2 13B have (I know that's silly but do the math)? Note that, as in [Transformer Math](#), LLaMA-3 has 3 big FFW matrices, two up-projection and one down-projection. We ignored the two "gating" einsum matrices in this section, but they behave the same as W_{in} in this section.

MVP
→ 3FD.L

Attention channel

Q heads → $W_Q \rightarrow DNH$

K heads → $W_K \rightarrow DKH$

V heads → $W_V \rightarrow DKH$

$W_O \rightarrow DNH$

$H = \frac{D}{N}$

$x L = \# \text{ layers}$

$2DNH + 2DKH = 2DH(N+k)$

Emb. layer + CM head

2DV

Question 2: Let's assume we're training with BS=16M tokens and using Adam. Ignoring parallelism for a moment, how much total memory is used by the model's parameters, optimizer state, and activations? Assume we store the parameters in bf16 and the optimizer state in fp32 and checkpoint activations three times per layer (after the three big FFW matmuls).

Ignore master weight.

$N \rightarrow$ per token

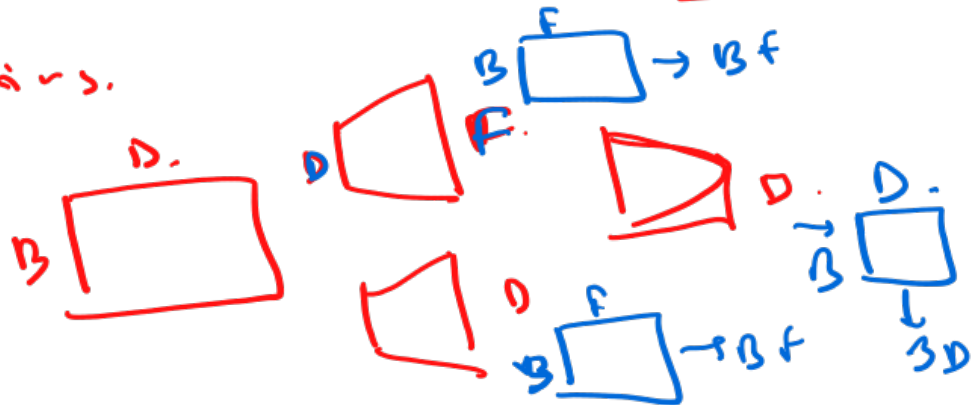
$2N \rightarrow$ bf16 weights

$FP32 \rightarrow 4N \rightarrow M$ (first moment Adam)

$4N \rightarrow V$
 $10N$

$$10 \times 13 \times 10^9 = 130 \text{ GB.}$$

Activations.



bf16.

$$2 \left(2BF + BD \right)$$

Question 3: Assume we want to train with 32k sequence length and a total batch size of 3M tokens on a TPUv5p 16x16x16 slice. Assume we want to use bfloat16 weights and a float32 optimizer, as above.

1. Can we use pure data parallelism? Why or why not? $130GB \rightarrow X$.
2. Can we use pure FSDP without being comms-bound? Why or why not? With pure FSDP, how much memory will be used per device (assume we do gradient checkpointing only after the 3 big FFW matrices). $2L(2Bf + Bv)$.
3. Can we use mixed FSDP + tensor parallelism? Why or why not? If so, what should X and Y be? How much memory will be stored per device? Using only roofline FLOPs estimates and ignoring attention, how long will each training step take at 40% MFU?

comms bound.

$$FSDP = \frac{B}{X} > \frac{C}{Wid.}$$

2550

$\frac{2550}{3} = 850$



$\frac{B}{X} > 850$

$$B > 850 \times 16 \times 16 \times 16 = 3.48M$$

$$130GB + 2L(2Bf + Bv)$$

$$+ 2LB(2f + v)$$

$$+ 2 \times 40 \times 3 \times 10^6 (13774 \times 2 + 512)$$

$$\frac{B}{N} > \frac{(C/w_{ic})^2}{M_x M_y F}$$

$$> \frac{2550^2}{M_x M_y F}$$



$$M_x M_y F \quad (2:1)$$

$$\begin{cases} 2 \rightarrow \text{FLOP} \\ 1 \rightarrow \text{TP} \end{cases}$$

$$F_{30P} \begin{bmatrix} \square & \square & \square & \square \\ \text{TP} & & & \end{bmatrix}$$

$$\frac{B}{N} > \frac{2550^2}{2F}$$

$$\frac{2550^2_{\text{total}}}{2 \times 13774} = < 3M$$

$$\frac{2550^2}{2F} \approx 235$$

total FLOPs - 2F and 4word.

(*) $\frac{F \text{ FLOPs}}{A \text{ calculate FLOPs}} = 6 \times \frac{3e6 \times 13e9}{4096 \times 4.59e14 \times 40/100}$

$X_{opt} = \sqrt{\frac{B}{F} N \frac{M_x}{M_y}} = \sqrt{\frac{3e6 \times 4096 \times 2}{13774 \times 1}}$

≈ 1333

024 → FLOP

$\left(\frac{4096}{13774}\right) \leftarrow 4 \rightarrow \text{TP}$

Question: How many FLOPs does LLaMA-3 perform per token per training step? This helps us determine how expensive the whole training process will be.

$$\begin{array}{r}
 \text{2 fwd} \leftarrow 6 \times 1 \times 70 \times 10^9 \quad \text{4 fwd} \cdot \\
 \hline
 4.2 \times 10^{14} \cdot \quad \cdot \quad 420 \times 10^9 \\
 \hline
 4.6 \times 10^{14} \cdot
 \end{array}$$

Question: LLaMA 3 was trained for about 15 trillion tokens. How many FLOPs is that total?

$$\begin{array}{r}
 \rightarrow 6 \times 15 \times 10^{12} \times 70 \times 10^9 \\
 \rightarrow \frac{6 \times 15 \times 10^{12} \times 70 \times 10^9}{4.6 \times 10^{14}}
 \end{array}$$

Question: Let's say we wanted to train on a full TPU v5p pod with $16 \times 20 \times 28 = 8960$ chips. How long would this take to train at 40% MFU in bfloat16, assuming we are compute-bound?

$$\frac{6 \times 15e^{12} \times 70e^9}{8960 \times \frac{40}{100} \times 4.6e^{14}} \text{ seconds.}$$

3096x

Question: LLaMA 3-70B was pretrained with a batch size of about 4M tokens. How many TPUs do we need at minimum to train with this batch size? You can assume bfloat16 parameters and float32 optimizer state, and that you checkpoint activations 2 times per layer (at the beginning of the layer and after attention).

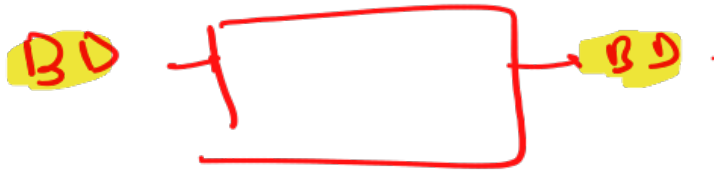
$N \rightarrow \text{params}$

$2N \rightarrow \text{weights}$

$4N \rightarrow m \text{ (optimizer state)}$

$4N \rightarrow v$

$$\frac{10N}{10^9} = 10 \times 70 \times 10^9 = (700 \text{ GB})$$



$2 \times 280 \text{ V}$
 $\downarrow$
 bfloat16

$$2 \times 2 \times 4 \times 10^6 \times 128 \times 14 \times 30$$

$$= 2 \times 8 \times 128 \times 14 \times 30 \times 10^7$$

$$= 2 \times 524.288 \times 10^7$$

$$\frac{1 \times 5.24 \times 10^{12}}{10^9} = 5.2 \times 10^3$$

$$= 5200 \text{ GB}$$

$$= (10.5 \text{ TB})$$

$$\frac{10.5 \times 700 \text{ GB}}{96 \text{ GB}}$$

#TPUs

Question: Under the same assumptions as the question above, if we use 8960 TPU v5p chips,
how much memory will we use per-chip?

action → per-chip + option

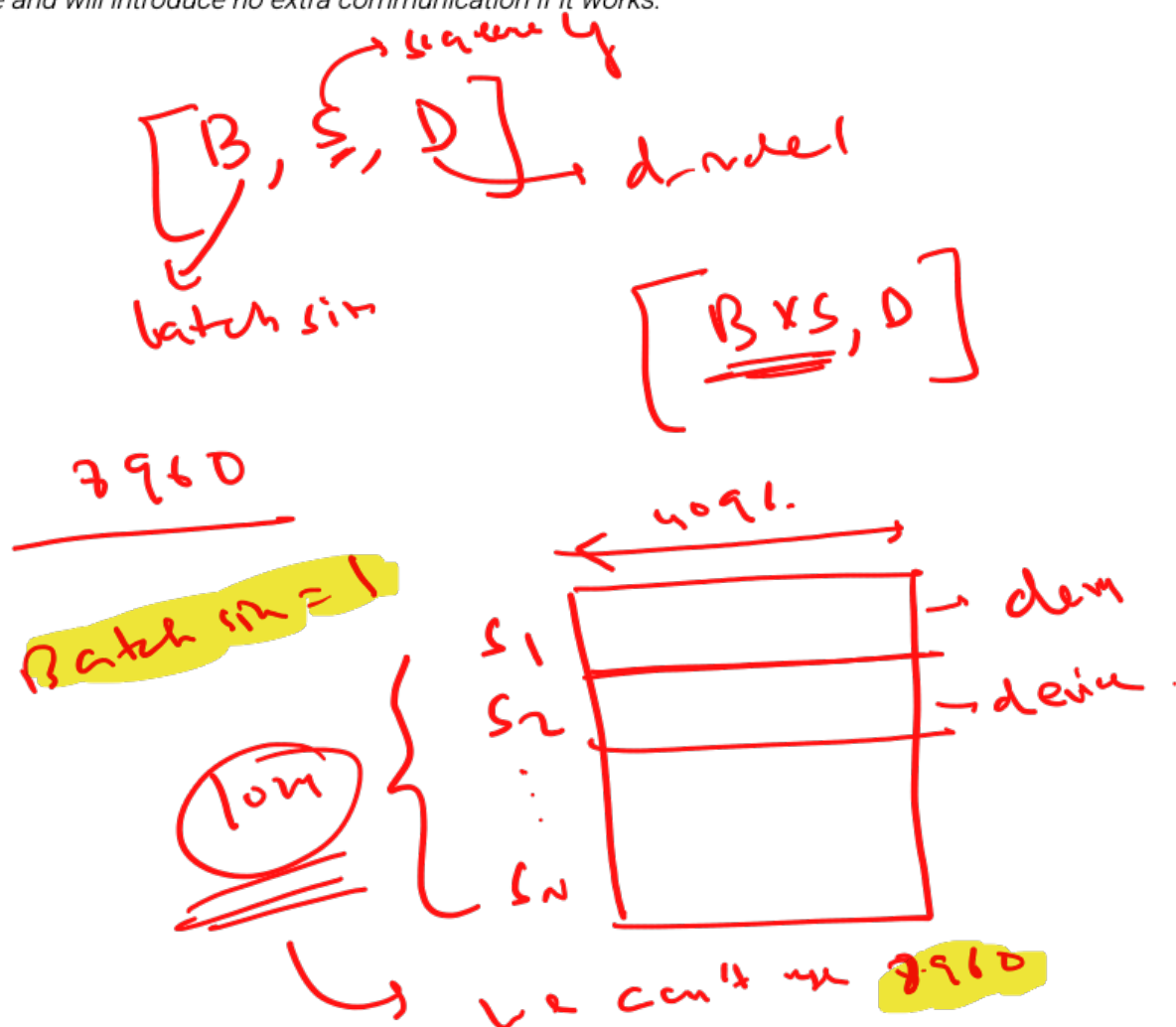
$$\frac{10.5 \times 700 \text{ GB}}{8960 \times 96 \times 10^9}$$

..

How to shard LLaMA 3-70B for training

Let's stick to our setting from above and say we want to train LLaMA 3-70B with 4M token batch size (1024 sequences of length 4096 per batch) on a TPU v5p pod of 8960 chips. Let's discuss what the best sharding strategy is for this model.

Question: Under the assumptions above, can we train our model with FSDP alone? To start, let's say we can't do any sequence/context parallelism. *This should be the first idea you have, since it's simple and will introduce no extra communication if it works.*



Question: Let's relax the requirement of not doing any sequence sharding. If we allow ourselves to do FSDP over both the batch *and* sequence axes, can we train LLaMA 3-70B with only FSDP on 8960 chips?

$$\frac{B}{X} > \frac{C}{w_{\text{ici}}}$$

$$\frac{n}{X} > 2500$$

per chip

$$\left[\begin{array}{c} B, S, D \\ \downarrow \text{FSDP} \end{array} \right]$$

FSDP.



$$B > \frac{2500}{3}$$

$$B > 833$$

$$\text{Batch size} = 4M$$

$$\frac{4 \times 10^6}{8960} \approx 445 \text{ chips}$$

$$> 800$$

$$468 < 850$$

Communication

Question: Now let's look at mixed tensor parallelism and FSDP. Does there exist some combination that lets us remain compute-bound? What amount of FSDP and tensor parallelism should we do if so?

$$\frac{B}{N} > \frac{(C/w_i)^2}{M \times M \times F}$$

$$B > \frac{w_i^2}{2 \times F}$$

$$B > 113$$

$$X_{opt} = \sqrt{\frac{B}{F} \cdot \frac{N \times M \times F}{M \times M}}$$

$$X_{up} = \sqrt{\frac{2 \times 7960 \times 4e^6}{28072}}$$

$$\approx 1613$$

$$\frac{2.049 \times 4}{t_{FSDP}} \quad t_{TP}$$